Chapter 4. The Fourier Transform

The class $\mathfrak{F}$, Fourier inversion and Poisson summation, the Paley–Wiener theorem

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Overview

For f on $\mathbb{R}$ with suitable regularity and decay, the Fourier transform and its inversion are

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x\xi}\, dx, \qquad f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi) e^{2\pi i x\xi}\, d\xi.$$

The theme of this chapter: the possibility of extending f to a holomorphic function is closely related to very rapid (e.g. exponential) decay of $\hat{f}$ at infinity. First, if f continues analytically to a horizontal strip and has moderate decrease, then $\hat{f}$ decays exponentially, and the inversion and Poisson summation formulas follow by contour integration. Second, assuming only Fourier inversion, the *Paley–Wiener theorem* says $\hat{f}$ is supported in $[-M, M]$ iff f extends to an entire function with $|f(z)| \le Ae^{2\pi M|z|}$ (*exponential type*). The decisive technique is shifting the contour (the real line) within a horizontal strip, exploiting that $e^{-2\pi i z\xi}$ decays or grows exponentially when $\operatorname{Im}(z) \neq 0$, according to the sign of $\xi \operatorname{Im}(z)$.

A function f is of **moderate decrease** if it is continuous and $|f(x)| \le A/(1+x^2)$ for some $A > 0$.

1 The class $\mathfrak{F}$

Book I proved Fourier inversion and Poisson summation when $|f(x)| \le \frac{A}{1+x^2}$ and $|\hat{f}(\xi)| \le \frac{A'}{1+\xi^2}$, a class containing e.g. the Poisson kernel $P_y(x) = \frac{1}{\pi}\frac{y}{y^2+x^2}$, $\hat{P}_y(\xi) = e^{-2\pi y|\xi|}$. Here we introduce a class suited to complex analysis.

Definition (the classes $\mathfrak{F}_a$ and $\mathfrak{F}$)

For $a > 0$, $\mathfrak{F}_a$ is the class of functions f such that:
(i) f is holomorphic in the horizontal strip $S_a = \{z \in \mathbb{C} : |\operatorname{Im}(z)| < a\}$;
(ii) there is $A > 0$ with $|f(x+iy)| \le \dfrac{A}{1+x^2}$ for all $x \in \mathbb{R}$ and $|y| < a$. $\mathfrak{F}$ is the class of functions belonging to $\mathfrak{F}_a$ for some a.

So $\mathfrak{F}_a$ consists of holomorphic functions on S_a of moderate decay on each horizontal line, uniformly in $-a < y < a$. Examples: $e^{-\pi z^2} \in \mathfrak{F}_a$ for all a; $\frac{1}{\pi}\frac{c}{c^2+z^2}$ (simple poles at $\pm ci$) belongs to $\mathfrak{F}_a$ for $0 < a < c$; $1/\cosh \pi z \in \mathfrak{F}_a$ for $|a| < 1/2$. By the Cauchy integral formulas, if $f \in \mathfrak{F}_a$ then $f^{(n)} \in \mathfrak{F}_b$ for $0 < b < a$. (The decrease $A/(1+x^2)$ may be weakened to $A/(1+|x|^{1+\epsilon})$ in most results below.)

2 Action of the Fourier transform on $\mathfrak{F}$

All three theorems are proved by contour integration.

Theorem 2.1

If f belongs to the class $\mathfrak{F}_a$ for some $a > 0$, then $|\hat{f}(\xi)| \le Be^{-2\pi b|\xi|}$ for any $0 \le b < a$.

Proof. $b = 0$ says $\hat{f}$ is bounded, clear from moderate decrease. Let $0 < b < a$ and $\xi > 0$. Shift the real line down by b: let $g(z) = f(z)e^{-2\pi i z\xi}$ and consider the rectangle with vertices $-R, R, R - ib, -R - ib$. The integral over the left vertical side is bounded by

$$\Big|\int_{-R-ib}^{-R} g(z)\, dz\Big| \le \int_0^b |f(-R-it)e^{-2\pi i(-R-it)\xi}|\, dt \le \int_0^b \frac{A}{R^2} e^{-2\pi t\xi}\, dt = O(1/R^2),$$

and similarly for the right side. By Cauchy's theorem, letting $R \to \infty$,

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - ib) e^{-2\pi i(x-ib)\xi}\, dx, \tag{1}$$

so $|\hat f(\xi)| \le \int_{-\infty}^{\infty} \frac{A}{1+x^2} e^{-2\pi b\xi}\,dx \le Be^{-2\pi b\xi}$. For $\xi < 0$ shift the real line up by b. ■

The further f extends (larger a), the better the decay of $\hat f$.

Theorem 2.2

If $f \in \mathfrak{F}$, then the Fourier inversion holds, namely

$$f(x) = \int_{-\infty}^{\infty} \hat f(\xi) e^{2\pi i x\xi}\,d\xi \qquad \text{for all } x \in \mathbb{R}.$$

Lemma 2.3

If A is positive and B is real, then $\int_0^\infty e^{-(A+iB)\xi}\,d\xi = \dfrac{1}{A+iB}$.

Proof. $|e^{-(A+iB)\xi}| = e^{-A\xi}$, so the integral converges, and $\int_0^R e^{-(A+iB)\xi}\,d\xi = \left[-\frac{e^{-(A+iB)\xi}}{A+iB}\right]_0^R \to \frac{1}{A+iB}$. ■

Proof of Theorem 2.2. Split $\int_{-\infty}^{\infty} \hat f(\xi) e^{2\pi i x\xi} d\xi = \int_{-\infty}^{0} + \int_0^{\infty}$. Say $f \in \mathfrak{F}_a$, $0 < b < a$. By (1) and the lemma (the integrals converge absolutely),

$$\begin{aligned}\int_0^\infty \hat f(\xi) e^{2\pi i x\xi}\,d\xi &= \int_0^\infty \int_{-\infty}^\infty f(u-ib) e^{-2\pi i(u-ib)\xi} e^{2\pi i x\xi}\,du\,d\xi = \int_{-\infty}^\infty f(u-ib) \int_0^\infty e^{-2\pi i(u-ib-x)\xi}\,d\xi\,du \\ &= \int_{-\infty}^\infty f(u-ib)\frac{1}{2\pi b + 2\pi i(u-x)}\,du = \frac{1}{2\pi i}\int_{-\infty}^\infty \frac{f(u-ib)}{u-ib-x}\,du = \frac{1}{2\pi i}\int_{L_1} \frac{f(\zeta)}{\zeta - x}\,d\zeta,\end{aligned}$$

$L_1 = \{u - ib : u \in \mathbb{R}\}$ traversed left to right. Similarly $\int_{-\infty}^0 \hat f(\xi) e^{2\pi i x\xi} d\xi = -\frac{1}{2\pi i}\int_{L_2} \frac{f(\zeta)}{\zeta - x}\,d\zeta$, L_2 the real line shifted up by b, left to right. Now for $x \in \mathbb{R}$ let γ_R be the rectangle with vertices $-R \pm ib$, $R \pm ib$. $f(\zeta)/(\zeta - x)$ has a simple pole at x with residue $f(x)$, so $f(x) = \frac{1}{2\pi i}\int_{\gamma_R} \frac{f(\zeta)}{\zeta - x}\,d\zeta$. As $R \to \infty$ the vertical sides contribute 0, and

$$f(x) = \frac{1}{2\pi i}\int_{L_1} \frac{f(\zeta)}{\zeta - x}\,d\zeta - \frac{1}{2\pi i}\int_{L_2} \frac{f(\zeta)}{\zeta - x}\,d\zeta = \int_0^\infty \hat f(\xi) e^{2\pi i x\xi}\,d\xi + \int_{-\infty}^0 \hat f(\xi) e^{2\pi i x\xi}\,d\xi.$$

■

Theorem 2.4 (Poisson summation formula)

If $f \in \mathfrak{F}$, then $\displaystyle\sum_{n\in\mathbb{Z}} f(n) = \sum_{n\in\mathbb{Z}} \hat f(n)$.

Proof. Say $f \in \mathfrak{F}_a$, $0 < b < a$. $1/(e^{2\pi i z} - 1)$ has simple poles with residue $1/(2\pi i)$ at the integers, so $f(z)/(e^{2\pi i z} - 1)$ has simple poles at the integers n with residues $f(n)/2\pi i$. Apply the residue formula to the rectangle γ_N with vertices $\pm(N + \frac12) \pm ib$:

$$\sum_{|n|\le N} f(n) = \int_{\gamma_N} \frac{f(z)}{e^{2\pi i z} - 1}\,dz.$$

Let $N \to \infty$; by moderate decrease the vertical sides contribute 0, and

$$\sum_{n\in\mathbb{Z}} f(n) = \int_{L_1} \frac{f(z)}{e^{2\pi i z} - 1}\,dz - \int_{L_2} \frac{f(z)}{e^{2\pi i z} - 1}\,dz, \tag{2}$$

L_1, L_2 the real line shifted down and up by b. If $|w| > 1$, $\frac{1}{w-1} = w^{-1}\sum_{n=0}^\infty w^{-n}$, so on L_1 (where $|e^{2\pi i z}| > 1$) $\frac{1}{e^{2\pi i z}-1} = e^{-2\pi i z}\sum_{n=0}^\infty e^{-2\pi i n z}$; if $|w| < 1$, $\frac{1}{w-1} = -\sum_{n=0}^\infty w^n$, so on L_2 $\frac{1}{e^{2\pi i z}-1} =$

$-\sum_{n=0}^{\infty} e^{2\pi i n z}$. Substituting in (2) and shifting L_1, L_2 back to the real line (by (1) and its analogue),

$$\sum_{n\in\mathbb{Z}} f(n) = \sum_{n=0}^{\infty} \int_{L_1} f(z)e^{-2\pi i(n+1)z}\,dz + \sum_{n=0}^{\infty} \int_{L_2} f(z)e^{2\pi i n z}\,dz = \sum_{n=0}^{\infty} \hat{f}(n+1) + \sum_{n=0}^{\infty} \hat{f}(-n) = \sum_{n\in\mathbb{Z}} \hat{f}(n).$$

■

Consequences. Since $e^{-\pi x^2}$ is its own Fourier transform (Example 1), for $t > 0$ and $a \in \mathbb{R}$ the change of variables $x \mapsto t^{1/2}(x + a)$ shows that $f(x) = e^{-\pi t(x+a)^2}$ has $\hat{f}(\xi) = t^{-1/2}e^{-\pi\xi^2/t}e^{2\pi i a\xi}$. Poisson summation gives

$$\sum_{n=-\infty}^{\infty} e^{-\pi t(n+a)^2} = \sum_{n=-\infty}^{\infty} t^{-1/2}e^{-\pi n^2/t}e^{2\pi i n a}. \tag{3}$$

For $a = 0$ this is the transformation law of the theta function $\vartheta(t) = \sum_{n=-\infty}^{\infty} e^{-\pi n^2 t}$ $(t > 0)$:

$$\vartheta(t) = t^{-1/2}\vartheta(1/t) \qquad \text{for } t > 0, \tag{4}$$

used in Chapter 6 for the functional equation of ζ; the general case is used in Chapter 10 for the Jacobi theta function. Likewise, since $1/\cosh \pi x$ is its own Fourier transform (Example 3), $f(x) = e^{-2\pi i a x}/\cosh(\pi x/t)$ has $\hat{f}(\xi) = t/\cosh(\pi(\xi + a)t)$, and

$$\sum_{n=-\infty}^{\infty} \frac{e^{-2\pi i a n}}{\cosh(\pi n/t)} = \sum_{n=-\infty}^{\infty} \frac{t}{\cosh(\pi(n + a)t)}, \tag{5}$$

used in Chapter 10 for the two-squares theorem.

3 Paley–Wiener theorem

Now we assume no analyticity, but the validity of Fourier inversion when $|f(x)| \le A/(1 + x^2)$ and $|\hat{f}(\xi)| \le A'/(1 + \xi^2)$ (proved in Book I). A partial converse to Theorem 2.1:

Theorem 3.1

Suppose $\hat{f}$ satisfies the decay condition $|\hat{f}(\xi)| \le Ae^{-2\pi a|\xi|}$ for some constants $a, A > 0$. Then $f(x)$ is the restriction to $\mathbb{R}$ of a function $f(z)$ holomorphic in the strip $S_b = \{z \in \mathbb{C} : |\mathrm{Im}(z)| < b\}$, for any $0 < b < a$.

Proof. Let $f_n(z) = \int_{-n}^{n} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$, entire by Theorem 5.4. For $z \in S_b$ define $f(z) = \int_{-\infty}^{\infty} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$; the integral converges absolutely, being majorized by $A\int e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi < \infty$ as $b < a$. Moreover for $z \in S_b$, $|f(z) - f_n(z)| \le A\int_{|\xi|\ge n} e^{-2\pi a|\xi|}e^{2\pi b|\xi|}\,d\xi \to 0$, so $f_n \to f$ uniformly in S_b, and f is holomorphic there by Theorem 5.2. ■

Corollary 3.2

If $\hat{f}(\xi) = O(e^{-2\pi a|\xi|})$ for some $a > 0$, and f vanishes in a non-empty open interval, then $f = 0$.

f is analytic in a region containing $\mathbb{R}$, so this follows from Theorem 4.8. In particular f and $\hat{f}$ cannot both have compact support unless $f = 0$.

Theorem 3.3 (Paley–Wiener)

Suppose f is continuous and of moderate decrease on $\mathbb{R}$. Then, f has an extension to the complex plane that is entire with $|f(z)| \le Ae^{2\pi M|z|}$ for some $A > 0$, if and only if $\hat{f}$ is supported in the interval $[-M, M]$.

Proof. *($\Leftarrow$)* If $\hat{f}$ is supported in $[-M, M]$, both f and $\hat{f}$ have moderate decrease and $f(x) = \int_{-M}^{M} \hat{f}(\xi)e^{2\pi i\xi x}\,d\xi$. Define $g(z) = \int_{-M}^{M} \hat{f}(\xi)e^{2\pi i\xi z}\,d\xi$; $g = f$ on $\mathbb{R}$, g is entire by Theorem 5.4, and for $z = x + iy$, $|g(z)| \le \int_{-M}^{M} |\hat{f}(\xi)|e^{-2\pi\xi y}\,d\xi \le Ae^{2\pi M|z|}$.

($\Rightarrow$) If $\hat{f}$ were supported in $[-M, M]$, the argument above would give the stronger bound $Ae^{2\pi M|y|}$. We reduce to that situation, adding also decay in x, and then remove the extra assumptions step by step.

Step 1. Assume f is entire with

$$|f(x + iy)| \le A\frac{e^{2\pi M|y|}}{1 + x^2}. \tag{6}$$

Then $\hat{f}(\xi) = 0$ for $|\xi| > M$. For $\xi > M$, shifting the real line down by $y > 0$ (as in (1)),

$$\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x - iy)e^{-2\pi i(x-iy)\xi}\,dx, \qquad |\hat{f}(\xi)| \le \int_{-\infty}^{\infty} A\frac{e^{2\pi My - 2\pi y\xi}}{1 + x^2}\,dx \le Ce^{-2\pi y(\xi - M)},$$

and letting $y \to \infty$ gives $\hat{f}(\xi) = 0$. For $\xi < -M$ shift up.

Step 2. Assume only

$$|f(x + iy)| \le Ae^{2\pi M|y|}. \tag{7}$$

Let $\xi > M$ and for $\epsilon > 0$ set $f_\epsilon(z) = \dfrac{f(z)}{(1 + i\epsilon z)^2}$. The factor $1/(1 + i\epsilon z)^2$ has absolute value ≤ 1 in the closed lower half-plane and tends to 1 as $\epsilon \to 0$; hence $\hat{f}_\epsilon(\xi) \to \hat{f}(\xi)$, since $|\hat{f}_\epsilon(\xi) - \hat{f}(\xi)| \le \int |f(x)||\frac{1}{(1+i\epsilon x)^2} - 1|\,dx \to 0$ by moderate decrease. Each f_ϵ satisfies $|f_\epsilon(x + iy)| \le A'\frac{e^{2\pi M|y|}}{1+x^2}$ in the lower half-plane, which is where Step 1 shifts the contour for $\xi > M$; so $\hat{f}_\epsilon(\xi) = 0$, and hence $\hat{f}(\xi) = 0$. For $\xi < -M$ argue in the upper half-plane with $1/(1 - i\epsilon z)^2$.

Step 3. It remains to show the hypotheses imply (7). Dividing by a constant, it suffices: if $|f(x)| \le 1$ for real x and $|f(z)| \le e^{2\pi M|z|}$ for all z, then $|f(x+iy)| \le e^{2\pi M|y|}$. This follows from the Phragmén–Lindelöf theorem below, rotated to the first quadrant $Q = \{x > 0, y > 0\}$: $F(z) = f(z)e^{2\pi iMz}$ is bounded by 1 on the positive real and imaginary axes and $|F(z)| \le Ce^{c|z|}$ in Q, so $|F(z)| \le 1$ in Q, i.e. $|f(z)| \le e^{2\pi My}$. The other quadrants are similar. ∎

The idea of Phragmén and Lindelöf adapts the maximum modulus principle to unbounded regions:

Theorem 3.4 (Phragmén–Lindelöf)

Suppose F is a holomorphic function in the sector $S = \{z : -\pi/4 < \arg z < \pi/4\}$ that is continuous on the closure of S. Assume $|F(z)| \le 1$ on the boundary of the sector, and that there are constants $C, c > 0$ such that $|F(z)| \le Ce^{c|z|}$ for all z in the sector. Then $|F(z)| \le 1$ for all $z \in S$.

Some growth restriction is needed: $F(z) = e^{z^2}$ is bounded by 1 on ∂S but unbounded on $\mathbb{R}_+$.

Proof. Subdue the "enemy" e^{z^2}: replace it by e^{z^α} with $\alpha < 2$; take $\alpha = 3/2$. For $\epsilon > 0$ let $F_\epsilon(z) = F(z)e^{-\epsilon z^{3/2}}$, with the principal branch, so $z^{3/2} = r^{3/2}e^{3i\theta/2}$ for $z = re^{i\theta}$, $-\pi < \theta < \pi$. F_ϵ is holomorphic in S and continuous up to the boundary, and $|e^{-\epsilon z^{3/2}}| = e^{-\epsilon r^{3/2}\cos(3\theta/2)}$. In the sector $-\pi/4 < \theta < \pi/4$ we have $-\frac{\pi}{2} < -\frac{3\pi}{8} < \frac{3\theta}{2} < \frac{3\pi}{8} < \frac{\pi}{2}$, so $\cos(3\theta/2)$ is bounded below by a positive constant. With $|F(z)| \le Ce^{c|z|}$, F_ϵ decreases rapidly as $|z| \to \infty$ in the closed sector and is bounded. Let $M = \sup_{z\in\overline{S}}|F_\epsilon(z)|$; if $F \not\equiv 0$, take w_j with $|F_\epsilon(w_j)| \to M$. Since $M \neq 0$ and F_ϵ

decays at infinity, the w_j accumulate at some $w \in \overline{S}$. By the maximum principle $w \notin S$, so $w \in \partial S$, where $|F| \leq 1$ and $|e^{-\epsilon z^{3/2}}| \leq 1$; so $M \leq 1$. Thus $|F_\epsilon| \leq 1$ on $\overline{S}$, and letting $\epsilon \to 0$ gives $|F| \leq 1$. ■

Another version of the Paley–Wiener idea characterizes Fourier transforms vanishing for $\xi < 0$:

Theorem 3.5

Suppose f and $\hat{f}$ have moderate decrease. Then $\hat{f}(\xi) = 0$ for all $\xi < 0$ if and only if f can be extended to a continuous and bounded function in the closed upper half-plane $\{z = x + iy : y \geq 0\}$ with f holomorphic in the interior.

Proof. If $\hat{f}(\xi) = 0$ for $\xi < 0$, then $f(x) = \int_0^\infty \hat{f}(\xi) e^{2\pi i \xi x}\, d\xi$, and we extend f for $y \geq 0$ by $f(z) = \int_0^\infty \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi$. The integral converges, and $|f(z)| \leq A \int_0^\infty \frac{d\xi}{1+\xi^2} < \infty$, so f is bounded. The functions $f_n(z) = \int_0^n \hat{f}(\xi) e^{2\pi i \xi z}\, d\xi$ converge to f uniformly in the closed half-plane, which gives continuity there and holomorphy in the interior.

Conversely, in the spirit of Theorem 3.3, for $\epsilon, \delta > 0$ set $f_{\epsilon,\delta}(z) = \dfrac{f(z + i\delta)}{(1 - i\epsilon z)^2}$, holomorphic in a region containing the closed upper half-plane. As before, by Cauchy's theorem (shifting the contour up), $\hat{f}_{\epsilon,\delta}(\xi) = 0$ for all $\xi < 0$. Passing to the limit successively, $\hat{f}_{\epsilon,0}(\xi) = 0$ for $\xi < 0$, and finally $\hat{f}(\xi) = \hat{f}_{0,0}(\xi) = 0$ for $\xi < 0$. ■

Remark

Compare Theorem 7.1: there, for a function holomorphic in a disc, the Fourier coefficients vanish for $n < 0$; here, for a function holomorphic in the upper half-plane, the Fourier transform vanishes for $\xi < 0$.